AR-Mini · How it is built: one engineer + one AI assistant + the physical world

AI proposes, a measurement decides. Every number in the CAD traces to calipers, a datasheet or a test. · 28.09 – 07.10.2026, real artefacts only

SYSTEM ARCHITECT · Teodora Kyoseva

Sets requirements · measures every part · prints, wires, solders, tests · decides what is true · signs off every version

owns the decisions | owns the measurements | owns the hardware | owns the commits



↓ *requirements, questions, photos* ↑ *measurements, test results*

1 · AI collaboration layer · Claude (one assistant, one long conversation)

Not autonomous agents: the same assistant takes five roles on request; nothing reaches the hardware without the architect.



Mechanical
CAD · kinematics

- AR-Mini_v1.scad v1.0 → v1.9
- parallelogram lift, 215° / 45°
- hull split for a 300 mm bed
- 3D model + collision found

output → review by architect



Electronics
power · signals

- architecture v1 + fuse tree
- XIAO breakout rev 0 (KiCad)
- carrier rev A spec
- 3 breadboard sheets

output → review by architect



Firmware
C++ · ESP32-S3

- bench_v1.ino (Mega)
- PlatformIO skeleton
- pin maps XIAO / DevKit-Lipo
- state machine plan

output → review by architect



Sourcing
BOM · suppliers

- BOM: 124 items with photos
- listing checks (thrusters!)
- BG shops vs Amazon.de
- what NOT to buy

output → review by architect

Documentation
log · portfolio

- FAILS_AND_FIXES: 142 entries
- LOG · TIMELINE · decisions
- GitHub repo structure
- posters + visual BOM

output → review by architect

↓ ↑

2 · Tools & ground truth

Where proposals become files, and where they are checked.

OpenSCAD · Fusion 360
parametric CAD, one file = whole robot

KiCad 9 + DRC
32 → 19 → 15 → clean (05.10)

Python
3D model, renders, posters, BOM

Datasheets & manuals
Olimex pinout, ROB0154 manual

Calipers SUW-DIGIP54
TD-8120MG 28.0 / 10 / 4.34 / 10.4

GitHub · ARmini
every version, honest dates

↓ ↑

3 · Physical reality

The only layer that cannot be argued with.



Creality K2 Pro
PLA fit prints → PETG

in use



Breadboard stands
Mega bench · XIAO · DevKit

in use from 07.10



UT61E+ · logic analyzer
volts, ohms, I²C, UART

in use



Thruster in a bucket
first wet test

next



Pool
draft, lift, rescue cycle

Oct – Dec

The loop · propose → build → measure → correct

Requirement (architect) → **Proposal** (AI: CAD / circuit / code) → **Build** (print · wire · solder) → **Measure** (calipers · meter · 3D view) → **Correct** (new version + log entry)

9 CAD versions in 9 days

Real corrections from this loop (all logged in FAILS_AND_FIXES.md)

- **3D view** — Cross-beams ran through the power box (v1.5–1.8). Box moved onto the beams → CAD v1.9.
- **DRC** — First PCB: 32 → 19 → 15 violations → clean, then ready for AISLER.
- **Arrival** — Thruster is 112 × 77 mm, mounted from the top → transom pad redesign (open).
- **Listing check** — Seven "thrusters" on Amazon were not thrusters → rule: thrust figure + ESC in the title.
- **Calipers** — TD-8120MG ear plane 28.0, holes 10 apart Ø4.34, shaft 10.4 → servo pocket fixed in v1.8.
- **Manual photos** — ROB0154 was assumed upside down for a day: motors high, not wet → hangers deleted.
- **Datasheet** — Olimex board uses GPIO5/6 internally → second I²C bus moved to GPIO8/9.

By the numbers

8 days, idea → orderable PCB

1.9 current CAD version

142 logged fails & fixes

124 BOM items, all with photos

3 DRC rounds to a clean board

0 numbers without a source

Rules that keep it honest

- AI output is a proposal until a measurement agrees.
- Every figure lives in the numbers register with its source.
- Failures are logged the day they happen, with the fix.
- The architect prints, wires, solders and tests.
- Commits carry the real date; earlier work is imported, not backdated.

Releases

- **R1 · Oct** — floats, lifts Pesho in the bathtub
- **R2 · Nov** — swims and steers in the pool (planned)
- **R3 · Dec** — full autonomous rescue cycle (planned)